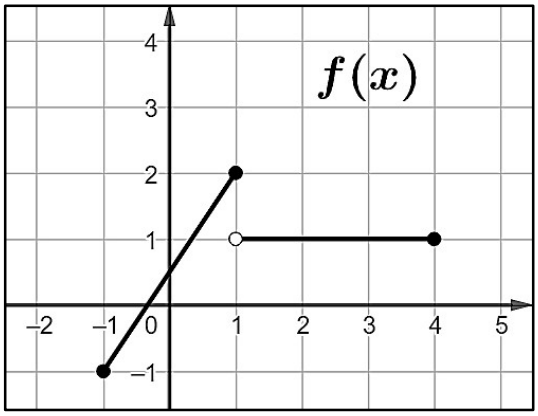
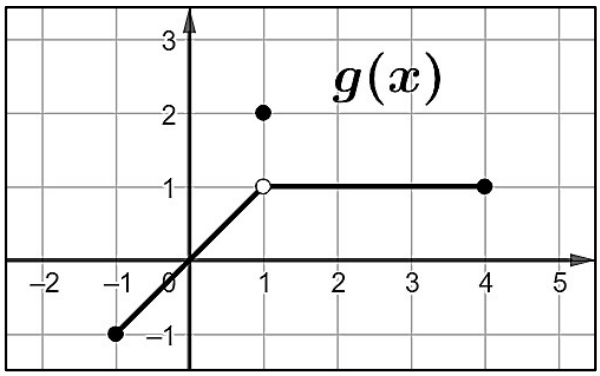
Worksheet 1-11 to 1-13 A Continuity #1
AP Calculus

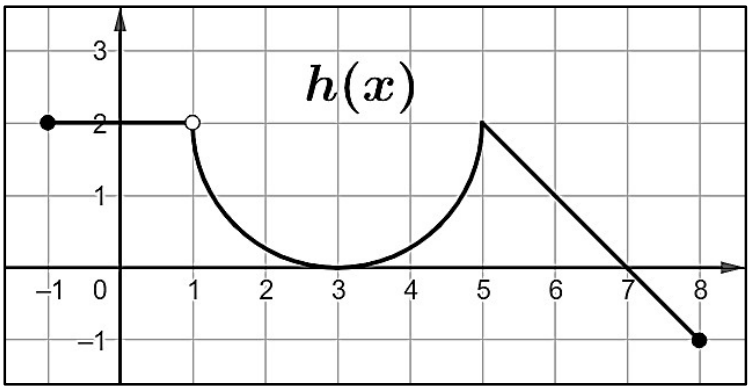
Name_____





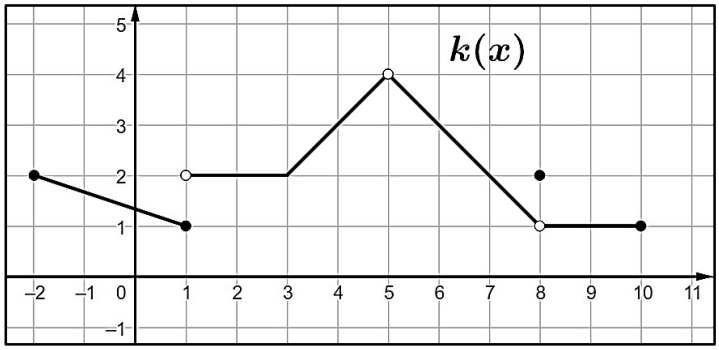
1. The function $f(x)$ is above. Is $f(x)$ continuous at $x = 1$? Explain your reasoning.

2. The function $g(x)$ is above. Is $g(x)$ continuous at $x = 1$? Explain your reasoning.



3. The function $h(x)$ is above. Is $h(x)$ continuous at $x = 5$? Explain your reasoning.

4. Is $h(x)$ continuous at $x = 1$? Explain your reasoning.



Directions: For 5-8, the function $k(x)$ is above. Use the graph of $k(x)$ to answer the following questions.

5. For $c = 8$, which of the following statement(s) are **FALSE**:

I. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x)$

II. $k(c)$ exists

III. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x) = k(c)$

A) I only B) II only C) III only D) I and II only E) I, II and III

---

6. For $c = 1$, which of the following statement(s) are **TRUE**:

I. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x)$

II. $k(c)$ exists

III. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x) = k(c)$

A) I only B) II only C) III only D) I and II only E) I, II and III

---

7. For $c = 5$, which of the following statement(s) are **TRUE**:

I. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x)$

II. $k(c)$ exists

III. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x) = k(c)$

A) I only B) II only C) III only D) I and II only E) I, II and III

---

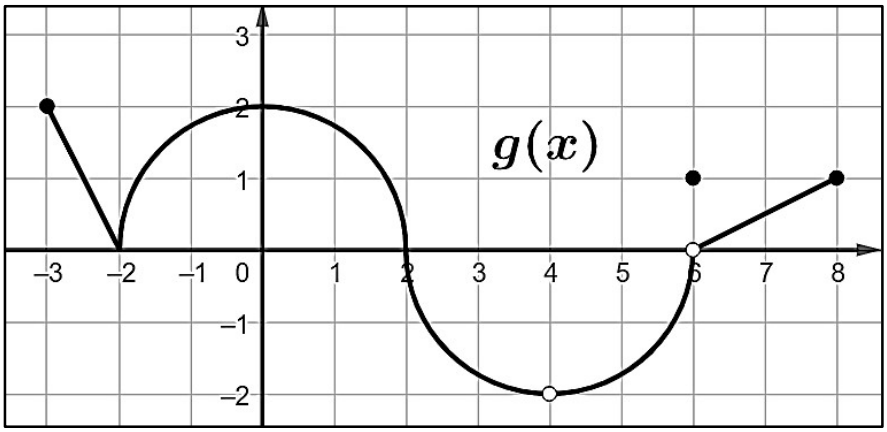
8. For $c = 3$, which of the following statement(s) are **TRUE**:

I. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x)$

II. $k(c)$ exists

III. $\lim_{x \to c^-} k(x) = \lim_{x \to c^+} k(x) = k(c)$

A) I only B) II only C) III only D) I and II only E) I, II and III



9. The graph of $g(x)$ is above. Use this graph to answer the following.

A) At what x value(s) is $g(x)$ NOT continuous? Classify each discontinuity and state which requirement for continuity was broken for each x value.

B) What is the domain of $g(x)$?

C) Is $g(x)$ continuous at $x = -2$? Explain using the definition of continuity at a point.

$$f(x) = \begin{cases} x - 3, & x < 0 \\ 2x + 1, & x \geq 0 \end{cases}$$

10. The piecewise function, $f(x)$, is given above. Is f continuous at $x = 0$? Explain your reasoning.

$$h(x) = \begin{cases} 3x - 1, & x < 2 \\ x^2 + 1, & x > 2 \end{cases}$$

11. The piecewise function, $h(x)$, is given above. Is h continuous at $x = 2$? Explain your reasoning.

$$k(x) = \begin{cases} 3x - 1, & x \leq -1 \\ \dfrac{x^2 - 2x - 3}{x + 1}, & x > -1 \end{cases}$$

12. The piecewise function, $k(x)$, is given above. Is k continuous at $x = -1$? Explain your reasoning.

$$g(x) = \begin{cases} 2x - 3, & x \leq 1 \\ x^2 + c, & x > 1 \end{cases}$$

13. The piecewise function, $g(x)$, is given above. Find the value of c such that $g(x)$ is continuous.

$$k(x) = \begin{cases} 2c - 3x, & x < -1 \\ c^2, & x \geq -1 \end{cases}$$

14. The piecewise function, $k(x)$, is given above. Find any value(s) of c such that $k(x)$ is continuous.

$$f(x) = \begin{cases} 3, & x < 0 \\ cx + d, & 0 \leq x \leq 4 \\ 2x - 13, & x > 4 \end{cases}$$

15. The piecewise function, $f(x)$, is given above. Find the values of c and d that make $f(x)$ is continuous.